

APM Project Management Qualification

Learner Study Pack



Budgeting and Cost Control

Introduction

This section of the Learner Study Pack will cover the Budget and Cost Control competence of the APM Project Management Qualification.

This section will enable you to gain an understanding of the different elements to consider when creating and monitoring project budgets.

Learning objective and outcomes

Learning objective:

22. Understand budgeting and cost control as the ability to estimate costs, develop and agree budgets and monitor actual costs against forecast costs.

By the end of this activity, you should be able to:

22a) Know how to create a budget (including the use of a Cost Breakdown Structure) and the different costs included in a budget (fixed, variable, direct, indirect etc.)

22b) Know how to forecast and refine budgets using cost control techniques e.g. earned value.

22c) Know how to monitor and report financial performance (including different types of financial reports).

22d) Know how to close down finances at the end of a project.

Study time

This section of the Learner Study Pack will take approximately 1 - 2 hours to complete.

Recommended resources

In addition to your Learner Study Pack, use the below references for more information on Budgeting and Cost Control:

- APM Body of Knowledge Seventh edition, 4.2.4 (Estimation); 4.2.8 (Cost planning); 4.3.1 (Progress monitoring and reporting)
- APM Learning Module, Budgeting and finance management
- APM Learning Handbook, Earned Value Management
Accessed at: <https://www.apm.org.uk/resources/apm-learning/>
- APM Resource, What is project cost planning and control?
Accessed at: <https://www.apm.org.uk/resources/what-is-project-management/what-is-project-cost-planning-and-control/>
- APM Resource, What is investment appraisal and project funding?
Accessed at: <https://www.apm.org.uk/resources/what-is-project-management/what-is-investment-appraisal-and-project-funding/>
- APM, Planning, Scheduling, Monitoring and Control - The Practical Project Management of Time, Cost and Risk
Accessed at: <https://www.apm.org.uk/resources/apm-learning/>
Accessed at: <https://www.apm.org.uk/book-shop/planning-scheduling-monitoring-and-control-the-practical-project-management-of-time-cost-and-risk/>

Introduction

Budgeting involves the estimation and allocation of financial resources required for project activities, while cost control focuses on monitoring and managing expenditures throughout the project lifecycle.

Together, these processes enable project managers to align financial resources with project goals, prevent budget overruns, and ensure optimal use of funds.

Robust budget and cost control mechanisms lay ensure delivery of projects on time and within the allocated financial parameters.

Creating a Budget

In the project profession, estimates are used for multiple purposes. One of which is to support budgeting and cost control. Budgets are created from estimates, looking at the approximation of time and cost to complete the scope of work to the required acceptance criteria.

Estimates are used:

- To enable budget setting and considerations of affordability.
- Supporting judgements to be made on value for money for the solution chosen.
- Estimates are an essential input into creating a resource schedule which then in turn provides the cost view of the project.

As part of creating a budget, consideration should be given to:

- The different types of costs involved and when they will hit the project to know when money will be needed to budget accordingly i.e. funding drawdowns at the right time to cover payment milestones.
- Information required from external sources, such as supplier costs, and how these align to the plan to enable budgeting.
- The assumptions being made on the project and how they are underpinning the cost estimates so they can be understood, validated and budgeted for.

Cost Breakdown Structure (CBS)

The Work Breakdown Structure (WBS) is used to create the Cost Breakdown Structure (CBS) by taking each of the work packages defined for the project and looking at the estimated costs related to the delivery of activity.

By doing this, a budget can be set for the project overall.

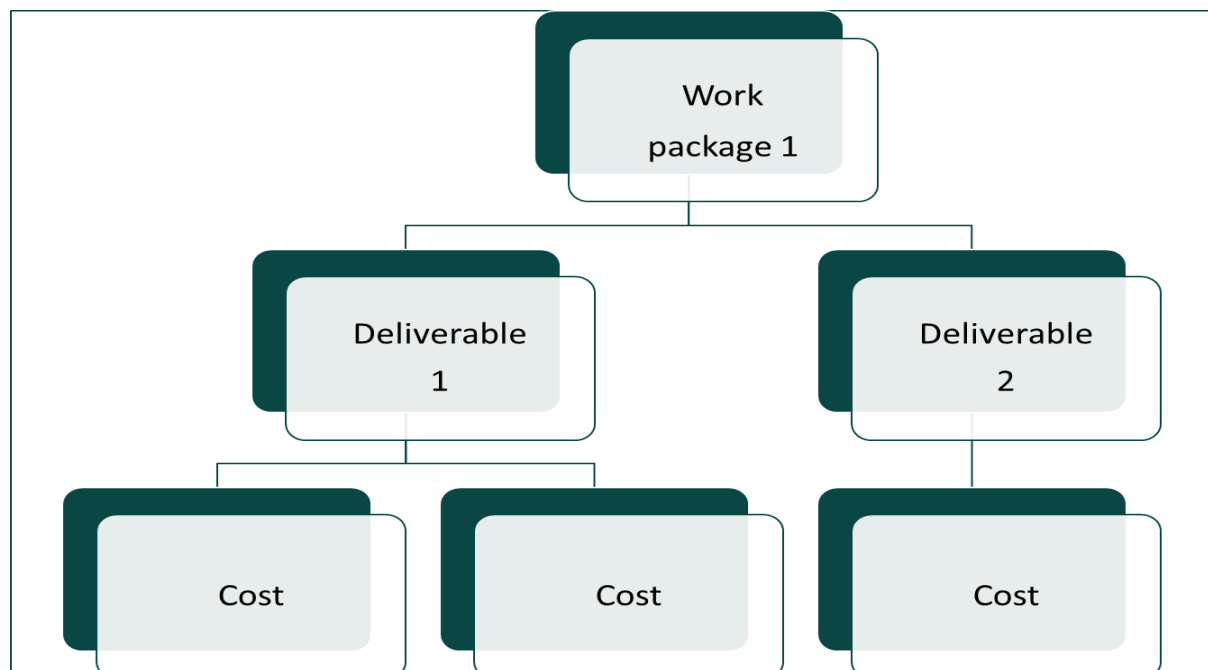
The CBS identifies all activity that has a cost associated with it, both labour and non-labour, in each work package.

These costs can then be collected, recorded, monitored and controlled as part of the financial reporting system.

Costs can be attributed to:

- People
- Equipment
- Materials
- Other resources required

The CBS will reflect the financial coding used in the project for booking and recording. This may be set at an organisational level as part of the organisation's financial reporting procedures.



Relationship between Different Costs

Knowing the overall cost of the project is a good start. However, this is not enough for project control.

What we need to understand is within that overall cost, where do the costs fall over time in the schedule? We are then able to manage resources, supplier payments and funding requests effectively.

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Project life cycle determines the frequency funds are to be released:

- In a linear life cycle, funds may only be released at decision gates when the costs spent to date are understood and the future forecasts have been approved
- Whereas in an iterative life cycle, the release of funds may be more frequent due to work being completed in short intervals.

This cost planning exercise is an essential part of the process to manage cashflow and ensure the project manager knows when funding drawdowns are needed.

Cost planning aspects include looking at the type of costs:

- **Are costs fixed?** For example, a supplier providing resources at a fixed price would remain this level irrespective of actual resource used by the supplier, and the monthly cost would be the same or fixed.
- **Or are they variable?** For example, materials that are needed regularly through the project to deliver the solution whereby the cost will vary dependent on what is needed or used. Therefore, the monthly cost could be different or variable.
- **Recurring costs**, for example, software licenses and maintenance which reoccur at set periods, happen periodically as an event happens and contribute multiple costs to the project at different times.
- **Or non-recurring**, for example, equipment set up costs, are a one-off cost to the project, happening once in a project life and contributing a single cost.
- It's important to consider whether they are direct or indirect costs as well, as this makes a difference to how easy it is to identify, track and manage them. Direct costs are directly related to the outputs of the project, whereas indirect costs are not, they are associated with the costs of operating the outputs for the business as a whole.

Cost Control Techniques

Cost control is necessary to identify variances to project budgets and cost forecasts to inform decision making.

Project managers use cost control techniques to:

- Minimise costs where possible.
- Identify areas of overspend.
- Correct unacceptable levels of overspend.
- Inform future projects by providing insight for lessons learnt.

Forecasts are created and used to determine project costs against scheduled activities.

When changes are made to the agreed baseline, it is important to update the cost information to refine budgets.

A recognised best practice approach to cost control is using earned value management which is ideal for identifying where a project is not progressing efficiently.

Earned value analysis reports on the project spend against the spend curve and the value of the work achieved. If the project is underspent or overspent against the planned spend line, knowing the value of work done at that point helps to refine budgets based on current project status.

The efficiency of spend is looked at through the cost performance indicator (CPI) and productivity through the schedule performance indicator (SPI).

A pictorial representation of project progress can be provided by tracking the Earned Value on an Earned Value graph.

Monitoring and Reporting Financial Performance

Progress monitoring enables meaningful reports to be shared to improve performance and enable decision making.

There are different approaches that can be taken to progress monitoring and reporting, the project manager and sponsor would agree appropriate methods to monitor financial performance early in the project.

Monitoring looks at how the project is performing against the agreed baseline.

Usual reporting could include:

- Actual costs vs forecasted costs.
- Actual spend vs forecasted spend against a set spend curve.
- Actual spend v actual work achieved looking at the efficiency of spend (earned value analysis).
- Estimated project budget and the forecasted cost overrun based on planned completion dates.
- Cashflow and drawdown requests.
- Resource costs.
- Change request costs.
- Contingency drawdown requests.
- Benefit realisation.

Monthly monitoring would be appropriate and typically a traffic light approach is used when reporting to flag the areas that are in or out of the agreed forecast including tolerance levels.

Financial performance is also monitored through different reviews that happen as part of project governance; decision gates, benefit reviews, stage reviews and audits will all monitor progress against baselined budgets, timelines, and success criteria.

Closing down Finances

When the project has completed delivery, as part of governance, the project sponsor will commence close down activities.

This will involve a project closure report, which is submitted to the final decision gate for endorsement.

As well as the report, the following activities need to take place:

Reassign Resources

- Back into the business, hand back over to the supplier, or exit from the organisation.
- To make sure not getting charged unnecessarily for resources that are no longer required.

Formally Close Contracts

- Facilitate closing all contracts in the supply chain.
- Making sure everything has been done, and everything expected to be received has been.

- Final accounts have been agreed.

Accounting and Management Systems Updated

- Close down of project on the accounting and management systems.
- Removing delegated authorities and project cost codes.
- No further money can be charged against the project.

Surplus Funding Returned

- Once all payments have been settled, any budget not spent needs returning to the business for use elsewhere.

Consideration should be given to how close down activities will happen early in the project life cycle, so as not to incur any additional costs that have not been factored into the budget.

Scenario One:

Robin, the Finance Manager, is always ready to 'lend a hand' and they would like the budgeting and cost control for the renovation project to be no different.

They are incredibly thorough, fastidious, and practical and have suggested that, as part of the project, they could review the budget that you have outlined to ensure that there are no surprises.

What are some of the factors that you should consider when creating your budget?

What information would you need to make your outlined budget more detailed?

Scenario Two:

As Robin has been given complete oversight over the budget of the project, they are starting to become concerned about the rate at which money is being spent and ensuring that the project's construction is good value for money.

They want to know how you're going to make sure that 'Snappy Appy' is getting the value of all of the money that is being spent.

What tools and techniques could you use to reassure Robin and all the stakeholders that you can demonstrate value for money during the project?

Given the potential impact, what report/s could you use to reassure Robin that they have visibility of the financial performance of the project?

What financial metrics could be used as part of the report?

Scenario Three:

When the project draws near to its completion, Robin wants to ensure that all funds have been accounted for, so that surplus can be returned to the business and used as part of the ongoing expansion of 'Snappy App'.

What steps would you take to close down the finances of the project?

Learning review questions

Before you attempt the exam practice questions in the next section, we recommend you review your learning to quickly see which aspects you don't understand or can't remember as well.

It will also boost your confidence by confirming what you have learnt and your demonstrated ability to apply that knowledge, so that you are ready to practise for the exam.

Here are some questions you may wish to answer to review your learning from the Budget and Cost Control section:

- What different project assumptions could have an impact on forecasted costs?
- How might governance reviews be used to understand financial performance?
- What are the benefits of tracking costs throughout the project?
- What are the different types of financial reporting and what are the different things that they tell you?
- Why is it important to close down finances at the end of a project?
- What are the different types of costs that may need to be considered in a project?
- What is the difference between a fixed and a variable cost?

Exam practice questions

22.1 You are the project manager for the construction of a new office building, you are tasked with creating a budget that includes design, materials, labour, technology and infrastructure. The project involves a diverse multidisciplinary team, and the success of the project relies heavily on effective budget management, where there is very little flexibility.

Which one of the following statements is correct when concerned with managing the budget and providing status evidence for stakeholders?
(1 mark)

- a. Categorise all costs, outlining the cost type of the various elements of the construction process, allowing for a more efficient analysis.
- b. Categorise all costs as fixed to simplify budget tracking, and evidence. The finance team will be able to determine any variances for stakeholder updates.
- c. Categorise all costs, wait until the project closes and adjust the budget reports based on project progress. This will clearly show the over or underspend.
- d. Categorise all material and staff costs as indirect costs to streamline budget calculations. This will allow for easier stakeholder reporting.

22.2 You have been asked to give a presentation to the Steering Group reviewing your project.

Which of the below metrics would be most useful to show costs committed and work achieved so far?
(1 mark)

- a. Planned Value.
- b. Actual Cost.
- c. Earned Value.
- d. Schedule Performance Index.

22.3 You are managing a project to build a new underground trainline for a client.

The initial budget for the project is £2 million.

As the project progresses, you notice that the actual costs are consistently under the budget, and there is a surplus of £150,000.

What would be an appropriate course of action for managing the surplus funds?
(1 mark)

- a. As the surplus is less than 10% of the total budget, do not report the surplus and keep it as a contingency for unexpected future expenses.
- b. Inform stakeholders of the financial performance using a routinely produced budget report.
- c. Produce a cost-benefit analysis to inform the client of the financial performance.
- d. Use a resource report to inform the stakeholders of the financial performance.

22.4 You have just completed the final day of scheduled work on your project, and the scope has been delivered as per the contract.

What is the first thing you do to start closing project finances?
(1 mark)

- a. Safely archive all contract documentation for governance and assurance records.
- b. Let stakeholders know that the project is complete and issue a project close report.
- c. Get the project sponsor to sign off that the scope is completed to the correct quality standard.
- d. Check there are no outstanding contract commitments including project overheads or standing costs.

22.5 You are the project manager for a project due to go into Definition Phase. Your task is to complete a project budget, including a Cost Breakdown Structure (CBS).

Explain:

- What is a Cost Breakdown Structure (CBS)? (1 mark)
- Four types of costs that could be included in a CBS. (4 marks)

22.6 Earned Value Management is a project management method that helps the project manager to determine the _____(a)_____

Earned Value Management can support many kinds of projects but it tends **not** to be very useful in _____(b)_____projects

(2 marks)

(a)

1. Value of realised benefits
2. Actual spend vs the budget plan
3. Value of the work done
4. Budgeted cost of work delivered

(b)

1. Long-term
2. Social impact
3. Construction
4. Linear life cycle

22.7 A key step in closing down project finances is to _____ **(a)** _____ to make sure that the project is not over-charged.

If there is any surplus budget at the end of a project, it should be _____ **(b)** _____.

(2 marks)

(a)

1. Reassign resources
2. Complete financial reports
3. Agree transition into BAU
4. Transfer sensitive financials

(b)

1. Spent before project end
2. Re-allocated to another project
3. Returned to the project sponsor
4. Returned to the business

22.8 State two review points during a project where it would be appropriate to report on financial performance.

(2 marks)

Exam practice answers

Question	Answer / Mark scheme
22.1	A
22.2	C
22.3	B
22.4	D
22.5	1 mark for correctly explaining a Cost Breakdown Structure – any of the below: • Identifies all activity within a project that has a cost associated with it. • Breaks down project activity into work packages and assigns costs to each work package. • Breaks down a high-level project budget into individual costs associated with the work required to produce deliverables. 1 mark for each correct explanation of a cost, 4 marks available in total. For example, but not limited to: • Fixed cost – supplied at a fixed price, no matter the level of resource used. • Variable cost – these vary according to the level of resource used. • Recurring cost – these recur at set periods within a project / are event-driven and recur throughout a project. • Non-recurring cost – these happen once in a project and contribute a single cost to the project. • Direct cost – these are directly related to the production of the project outputs. • Indirect cost – these are associated with the cost of operating the business, not specific to the project.
22.6	(a) 3 (b) 2
22.7	(a) 1 (b) 4
22.8	1 mark for each review, 2 marks available in total, for example: • Decision gates • Benefit reviews • Stage reviews • Audits • Post-implementation reviews